Scripting with PHP-II — Complete Practical Solutions

Program 1 — Temperature Difference

Question: Read 6 AM and 6 PM temperatures using an HTML form. Find the absolute difference, check whether both are integers, and round the first temperature to 2 decimal places.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Temperature</title>
</head>
<body>

<h2>Temperature Calculation</h2>

<form method="post" action="temperature.php">

    <label>Temperature at 6 AM:</label>
    <input type="text" name="temp1" required>
    <br><br>

    <label>Temperature at 6 PM:</label>
    <input type="text" name="temp2" required>
    <br><br>

    <input type="submit" value="Calculate">

</form>

</body>
</html>
```

temperature.php

```
<?php

$temp1 = $_POST['temp1'];
$temp2 = $_POST['temp2'];

// Absolute difference
$difference = abs($temp1 - $temp2);

// Check whether temperatures are integers
if (filter_var($temp1, FILTER_VALIDATE_INT) !== false) {
    $check1 = "6 AM temperature is an integer.";
} else {
    $check1 = "6 AM temperature is not an integer.";
```

```
}

if (filter_var($temp2, FILTER_VALIDATE_INT) !== false) {
    $check2 = "6 PM temperature is an integer.";
} else {
    $check2 = "6 PM temperature is not an integer.";
}

// Round first temperature to 2 decimal places
$roundedTemp1 = round($temp1, 2);

echo "<h2>Temperature Result</h2>";

echo "6 AM Temperature: " . $temp1 . "<br>";
echo "6 PM Temperature: " . $temp2 . "<br>";
echo "Absolute Difference: " . $difference . "<br>";
echo $check1 . "<br>";
echo $check2 . "<br>";
echo "6 AM Temperature after rounding: " . $roundedTemp1;

?>
```

Program 2 — Birthdate Formatting

The journal asks to read the birthdate and print it in different formats using PHP date format specifiers.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Birthdate</title>
</head>
<body>

<h2>Birthdate Format</h2>

<form method="post" action="birthdate.php">

    <label>Enter Birthdate:</label>
    <input type="date" name="birthdate" required>

    <br><br>

    <input type="submit" value="Display Date">

</form>

</body>
</html>
```

birthdate.php

```
<?php

$birthdate = $_POST['birthdate'];

$date = new DateTime($birthdate);

echo "<h2>Birthdate in Different Formats</h2>";

echo "Format 1 (d-m-Y): " . $date->format("d-m-Y") . "<br>";
echo "Format 2 (d/m/Y): " . $date->format("d/m/Y") . "<br>";
echo "Format 3 (d M Y): " . $date->format("d M Y") . "<br>";
echo "Format 4 (d F Y): " . $date->format("d F Y") . "<br>";
echo "Format 5 (l, d F Y): " . $date->format("l, d F Y") . "<br>";
echo "Format 6 (Y-m-d): " . $date->format("Y-m-d") . "<br>";

?>
```

Important Date Specifiers

Specifier	Meaning
`d`	Day
`m`	Month number
`Y`	4-digit year
`y`	2-digit year
`M`	Short month name
`F`	Full month name
`l`	Full day name

Program 3 — College Name String Operations

The program must count characters, check whether the last word is **college** case-insensitively, and replace `college` with `university`.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>College Name</title>
</head>
<body>

<h2>College Name</h2>
```

```
<form method="post" action="college.php">

    <label>Enter College Name:</label>
    <input type="text" name="college" required>

    <br><br>

    <input type="submit" value="Submit">

</form>

</body>
</html>
```

college.php

```
<?php

$college = trim($_POST['college']);

// Number of characters
$characters = strlen($college);

// Convert string into words
$words = preg_split('/\s+/', $college);

// Get last word
$lastWord = end($words);

echo "<h2>College Details</h2>";

echo "College Name: " . $college . "<br>";
echo "Number of Characters: " . $characters . "<br>";

// Case-insensitive check
if (strcasecmp($lastWord, "college") == 0) {

    echo "Last word is 'college'.<br>";

    // Replace last occurrence of college
    $newCollege = preg_replace(
        '/college$/i',
        'university',
        $college
    );

    echo "After Replacement: " . $newCollege;

} else {

    echo "Last word is not 'college'.<br>";
    echo "No replacement required.";

}
```

```
?>
```

Program 4 — Cookie

The requirement is to remember the username even after the browser is closed and reopened. The cookie is valid for **1 day**.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Cookie Example</title>
</head>
<body>

<h2>Login Form</h2>

<form method="post" action="first.php">

    <label>Username:</label>
    <input type="text" name="username" required>

    <br><br>

    <label>Password:</label>
    <input type="password" name="password" required>

    <br><br>

    <input type="submit" value="Login">

</form>

</body>
</html>
```

first.php

```
<?php

$username = $_POST['username'];

// Cookie valid for 1 day
setcookie("username", $username, time() + (24 * 60 * 60), "/");

echo "<h2>Hello " . htmlspecialchars($username) . "</h2>";

echo "<p>Your username has been stored in a cookie for 1 day.</p>";

echo '<a href="second.php">Check Cookie</a>';
```

```
?>
```

second.php

```
<?php

if (isset($_COOKIE['username'])) {

    $username = $_COOKIE['username'];

    echo "<h2>Hello " . htmlspecialchars($username) .
         ", your cookie is still active!</h2>";

} else {

    echo "<h2>Cookie has expired or does not exist.</h2>";

}

?>
```

How to demonstrate

1. Open `index.html`.
2. Enter username and password.
3. Submit the form.
4. Open `second.php`.
5. You will see:

Hello Mehul, your cookie is still active!

6. Close and reopen the browser.
7. Open `second.php` again.
8. The username should still be available.
9. After 1 day, the cookie expires.

Program 5 — Session

The session should remember the username while the browser session is active and should not permanently store it like a 1-day cookie.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Session Example</title>
</head>
<body>
```

```
<h2>Login Form</h2>

<form method="post" action="first.php">

    <label>Username:</label>
    <input type="text" name="username" required>

    <br><br>

    <label>Password:</label>
    <input type="password" name="password" required>

    <br><br>

    <input type="submit" value="Login">

</form>

</body>
</html>
```

first.php

```
<?php

session_start();

$username = $_POST['username'];

$_SESSION['username'] = $username;

echo "<h2>Hello " . htmlspecialchars($username) . "</h2>";

echo "<p>Your session has been started.</p>";

echo '<a href="second.php">Check Session</a>';

?>
```

second.php

```
<?php

session_start();

if (isset($_SESSION['username'])) {

    $username = $_SESSION['username'];

    echo "<h2>Hello " . htmlspecialchars($username) .
         ", your session is active!</h2>";

} else {

    echo "<h2>Session is not defined.</h2>";
```

```
}

?>
```

Important difference

Cookie:

```
setcookie();
```

Session:

```
session_start();
$_SESSION['username'];
```

Program 6 — Text File Handling

The journal requires a text file `log.txt` with operations for append, display/search, and delete.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Visitor Log</title>
</head>
<body>

<h2>Visitor Log</h2>

<form method="post" action="first.php">

    <label>Visitor Name:</label>
    <input type="text" name="name">

    <br><br>

    <label>Select Operation:</label>

    <select name="operation">

        <option value="append">Append</option>
        <option value="display">Display</option>
        <option value="delete">Delete File</option>

    </select>

    <br><br>

    <input type="submit" value="Submit">
```

```
</form>

</body>
</html>
```

first.php

```
<?php

$file = "log.txt";

$operation = $_POST['operation'];

if ($operation == "append") {

    $name = trim($_POST['name']);

    if ($name != "") {

        // Open file in append mode
        $fp = fopen($file, "a");

        // Write visitor name
        fwrite($fp, $name . PHP_EOL);

        fclose($fp);

        echo "<h3>Visitor name added successfully.</h3>";

    } else {

        echo "<h3>Please enter visitor name.</h3>";

    }

} elseif ($operation == "display") {

    if (file_exists($file)) {

        $fp = fopen($file, "r");

        echo "<h2>Visitor List</h2>";

        echo "<table border='1' cellpadding='10'>";
        echo "<tr><th>No.</th><th>Visitor Name</th></tr>";

        $count = 1;

        while (($line = fgets($fp)) !== false) {

            echo "<tr>";
            echo "<td>" . $count . "</td>";
            echo "<td>" . htmlspecialchars(trim($line)) . "</td>";
            echo "</tr>";

            $count++;
```

```
        }
        echo "</table>";
        fclose($fp);
    } else {
        echo "<h3>Log file does not exist.</h3>";
    }
} elseif ($operation == "delete") {
    if (file_exists($file)) {
        unlink($file);
        echo "<h3>Log file deleted successfully.</h3>";
    } else {
        echo "<h3>Log file does not exist.</h3>";
    }
}
?>
```

File modes used

Mode	Meaning
`r`	Read
`w`	Write and overwrite
`a`	Append without deleting existing data

Program 7 — Profile Picture Upload

The program requires enrolment number and profile picture, with the picture renamed using the enrolment number and stored in the `htdocs/images` folder.

Folder Structure

For XAMPP:

```
htdocs
|
```

```
├── program7
│   ├── index.html
│   └── upload.php
│
└── images
```

Create the `images` folder inside `htdocs`.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Profile Picture Upload</title>
</head>
<body>

<h2>Student Profile Picture</h2>

<form method="post"
      action="upload.php"
      enctype="multipart/form-data">

    <label>Enrolment Number:</label>
    <input type="text" name="enrolment" required>

    <br><br>

    <label>Profile Picture:</label>
    <input type="file"
           name="profile"
           accept="image/*"
           required>

    <br><br>

    <input type="submit" value="Upload">

</form>

</body>
</html>
```

upload.php

```
<?php

$enrolment = $_POST['enrolment'];

$fileName = $_FILES['profile']['name'];
$fileTmpName = $_FILES['profile']['tmp_name'];
$fileError = $_FILES['profile']['error'];

if ($fileError == 0) {
```

```
    // Get original file extension
    $extension = pathinfo($fileName, PATHINFO_EXTENSION);

    // New file name
    $newFileName = $enrolment . "." . $extension;

    // Destination folder
    $destination = "C:/xampp/htdocs/images/" . $newFileName;

    // Move uploaded file
    if (move_uploaded_file($fileTmpName, $destination)) {

        echo "<h2>Profile picture uploaded successfully.</h2>";

        echo "Enrolment Number: " .
             htmlspecialchars($enrolment) . "<br>";

        echo "File Name: " .
             htmlspecialchars($newFileName);

    } else {

        echo "<h2>File upload failed.</h2>";

    }

} else {

    echo "<h2>Error while uploading file.</h2>";

}

?>
```

Important Form Attribute

For file upload, this is compulsory:

```
enctype="multipart/form-data"
```

Program 8 — Student Result Using OOP

The journal specifically requires a `Student` class with public properties, constructor, private `calc()` and `res()` methods, private result properties, and public `disp()` method.

student.php

```
<?php

class Student
{
```

```
// Public properties
public $enrolment_no;
public $name;
public $marks;

// Private properties
private $total;
private $perc;
private $result;

// Constructor
public function __construct($enrolment_no, $name, $marks)
{
    $this->enrolment_no = $enrolment_no;
    $this->name = $name;
    $this->marks = $marks;
}

// Calculate total and percentage
private function calc()
{
    $this->total = array_sum($this->marks);

    $this->perc = $this->total / 5;
}

// Calculate result
private function res()
{
    $this->result = "PASS";

    // If any subject mark is less than 35
    foreach ($this->marks as $mark) {

        if ($mark < 35) {

            $this->result = "FAIL";
            break;

        }
    }
}

// Display student information
public function disp()
{
    // Call private methods
    $this->calc();
    $this->res();

    echo "<h2>Student Result</h2>";

    echo "<table border='1' cellpadding='10'>";

    echo "<tr>";
    echo "<th>Enrolment No.</th>";
    echo "<td>" . htmlspecialchars($this->enrolment_no) . "</td>";
```

```
        echo "</tr>";

        echo "<tr>";
        echo "<th>Name</th>";
        echo "<td>" . htmlspecialchars($this->name) . "</td>";
        echo "</tr>";

        echo "<tr>";
        echo "<th>Subject 1</th>";
        echo "<td>" . $this->marks[0] . "</td>";
        echo "</tr>";

        echo "<tr>";
        echo "<th>Subject 2</th>";
        echo "<td>" . $this->marks[1] . "</td>";
        echo "</tr>";

        echo "<tr>";
        echo "<th>Subject 3</th>";
        echo "<td>" . $this->marks[2] . "</td>";
        echo "</tr>";

        echo "<tr>";
        echo "<th>Subject 4</th>";
        echo "<td>" . $this->marks[3] . "</td>";
        echo "</tr>";

        echo "<tr>";
        echo "<th>Subject 5</th>";
        echo "<td>" . $this->marks[4] . "</td>";
        echo "</tr>";

        echo "<tr>";
        echo "<th>Total</th>";
        echo "<td>" . $this->total . "</td>";
        echo "</tr>";

        echo "<tr>";
        echo "<th>Percentage</th>";
        echo "<td>" . $this->perc . "%</td>";
        echo "</tr>";

        echo "<tr>";
        echo "<th>Result</th>";
        echo "<td>" . $this->result . "</td>";
        echo "</tr>";

        echo "</table>";
    }
}


// Create object
$student = new Student(
    "SPU001",
    "Rahul Patel",
    array(78, 85, 69, 72, 88)
```

```
);

// Display result
$student->disp();

?>
```

OOP concepts used

- Class
- Object
- Public properties
- Private properties
- Constructor
- Private methods
- Public method
- Encapsulation

Program 9 — Employee Database

The journal requires an `employee` table in the **LYBCA database** with `Empid`, `Ename`, `Deptcode`, `Joining date`, and `Gender`.

Step 1 — Create Database

Run this in **phpMyAdmin → SQL**:

```
CREATE DATABASE LYBCA;
```

Then select `LYBCA` and run:

```
CREATE TABLE employee
(
    Empid INT(3) PRIMARY KEY,
    Ename VARCHAR(35),
    Deptcode CHAR(1),
    JoiningDate DATE,
    Gender CHAR(1)
);
```

Department Codes

```
A = Admin
S = Sales
F = Finance
```

Gender Codes

```
M = Male
F = Female
```

Common Database Connection

Create:

db.php

```
<?php

$conn = mysqli_connect(
    "localhost",
    "root",
    "",
    "LYBCA"
);

if (!$conn) {

    die("Database connection failed: " . mysqli_connect_error());

}

?>
```

If your MySQL password is different, replace the empty password `""`.

9(A) Insert Employee Record

The journal requires an HTML form to read employee details and insert them into the table.

insert.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Insert Employee</title>
</head>
<body>

<h2>Employee Registration</h2>

<form method="post" action="insert.php">

    Employee ID:
    <input type="text" name="empid" required>
    <br><br>

    Employee Name:
    <input type="text" name="ename" required>
```

```
    <br><br>

    Department:
    <input type="radio" name="deptcode" value="A"> Admin
    <input type="radio" name="deptcode" value="S"> Sales
    <input type="radio" name="deptcode" value="F"> Finance
    <br><br>

    Joining Date:
    <input type="date" name="joiningdate" required>
    <br><br>

    Gender:
    <select name="gender">

        <option value="M">Male</option>
        <option value="F">Female</option>

    </select>

    <br><br>

    <input type="submit" value="Insert">

</form>

</body>
</html>
```

insert.php

```
<?php

include "db.php";

$empid = $_POST['empid'];
$ename = $_POST['ename'];
$deptcode = $_POST['deptcode'];
$joiningdate = $_POST['joiningdate'];
$gender = $_POST['gender'];

$sql = "INSERT INTO employee
        VALUES
        ('$empid', '$ename', '$deptcode', '$joiningdate', '$gender')";

if (mysqli_query($conn, $sql)) {

    echo "<h3>Employee record inserted successfully.</h3>";

} else {

    echo "Error: " . mysqli_error($conn);

}

mysqli_close($conn);
```

```
?>
```

9(B) Check Whether Employee Exists

The requirement is to read `empid` and display **Record exists** or **Record does not exist**.

check.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Check Employee</title>
</head>
<body>

<h2>Check Employee Record</h2>

<form method="post" action="check.php">

    Employee ID:
    <input type="text" name="empid" required>

    <br><br>

    <input type="submit" value="Check">

</form>

</body>
</html>
```

check.php

```
<?php

include "db.php";

$empid = $_POST['empid'];

$sql = "SELECT * FROM employee WHERE Empid = '$empid'";

$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {

    echo "<h3>Record exists</h3>";

} else {

    echo "<h3>Record does not exist</h3>";
```

```
}

mysqli_close($conn);

?>
```

9(C) Insert if New / Update if Existing

The question requires displaying the record **before updating and after updating**.

update.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Insert or Update</title>
</head>
<body>

<h2>Employee Details</h2>

<form method="post" action="update.php">

    Employee ID:
    <input type="text" name="empid" required>
    <br><br>

    Employee Name:
    <input type="text" name="ename" required>
    <br><br>

    Department:

    <input type="radio" name="deptcode" value="A"> Admin
    <input type="radio" name="deptcode" value="S"> Sales
    <input type="radio" name="deptcode" value="F"> Finance

    <br><br>

    Joining Date:
    <input type="date" name="joiningdate" required>
    <br><br>

    Gender:

    <select name="gender">
        <option value="M">Male</option>
        <option value="F">Female</option>
    </select>

    <br><br>

    <input type="submit" value="Submit">
```

```
</form>

</body>
</html>
```

update.php

```
<?php

include "db.php";

$empid = $_POST['empid'];
$ename = $_POST['ename'];
$deptcode = $_POST['deptcode'];
$joiningdate = $_POST['joiningdate'];
$gender = $_POST['gender'];

// Check record
$sql = "SELECT * FROM employee WHERE Empid='$empid'";

$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {

    // Display BEFORE
    $row = mysqli_fetch_assoc($result);

    echo "<h2>Record Before Updating</h2>";

    echo "Employee ID: " . $row['Empid'] . "<br>";
    echo "Name: " . $row['Ename'] . "<br>";
    echo "Department: " . $row['Deptcode'] . "<br>";
    echo "Joining Date: " . $row['JoiningDate'] . "<br>";
    echo "Gender: " . $row['Gender'] . "<br><br>";

    // Update
    $sql = "UPDATE employee SET
            Ename='$ename',
            Deptcode='$deptcode',
            JoiningDate='$joiningdate',
            Gender='$gender'
            WHERE Empid='$empid'";

    mysqli_query($conn, $sql);

    // Display AFTER
    $sql = "SELECT * FROM employee WHERE Empid='$empid'";

    $result = mysqli_query($conn, $sql);

    $row = mysqli_fetch_assoc($result);

    echo "<h2>Record After Updating</h2>";

    echo "Employee ID: " . $row['Empid'] . "<br>";
```

```
    echo "Name: " . $row['Ename'] . "<br>";
    echo "Department: " . $row['Deptcode'] . "<br>";
    echo "Joining Date: " . $row['JoiningDate'] . "<br>";
    echo "Gender: " . $row['Gender'] . "<br>";

} else {

    // Insert new record
    $sql = "INSERT INTO employee
            VALUES
            ('$empid','$ename','$deptcode','$joiningdate','$gender')";

    mysqli_query($conn, $sql);

    echo "<h3>New record inserted successfully.</h3>";
}

mysqli_close($conn);

?>
```

9(D) Display First Record — Form Layout

The question asks to fetch the first employee record and display it in form layout.

firstrecord.php

```
<?php

include "db.php";

$sql = "SELECT * FROM employee LIMIT 1";

$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {

    $row = mysqli_fetch_assoc($result);

    echo "<h2>First Employee Record</h2>";

    echo "Employee ID: " . $row['Empid'] . "<br><br>";

    echo "Employee Name: " .
         $row['Ename'] . "<br><br>";

    echo "Department: " .
         $row['Deptcode'] . "<br><br>";

    echo "Joining Date: " .
         $row['JoiningDate'] . "<br><br>";

    echo "Gender: " .
```

```
        $row['Gender'];

} else {

    echo "No records found.";

}

?>
```

9(E) Display First Record — Table Layout

The required output is in tabular format.

firsttable.php

```
<?php

include "db.php";

$sql = "SELECT * FROM employee LIMIT 1";

$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {

    $row = mysqli_fetch_assoc($result);

    echo "<h2>First Employee Record</h2>";

    echo "<table border='1' cellpadding='10'>";

    echo "<tr>";
    echo "<th>Employee ID</th>";
    echo "<th>Employee Name</th>";
    echo "<th>Department</th>";
    echo "<th>Joining Date</th>";
    echo "<th>Gender</th>";
    echo "</tr>";

    echo "<tr>";

    echo "<td>" . $row['Empid'] . "</td>";
    echo "<td>" . $row['Ename'] . "</td>";
    echo "<td>" . $row['Deptcode'] . "</td>";
    echo "<td>" . $row['JoiningDate'] . "</td>";
    echo "<td>" . $row['Gender'] . "</td>";

    echo "</tr>";

    echo "</table>";

} else {
```

```
    echo "No records found.";

}

?>
```

9(F) Display Employees by Gender

The question requires reading gender, displaying all employees of that gender, and printing the total number.

gender.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Employee Gender</title>
</head>
<body>

<h2>Employee Details by Gender</h2>

<form method="post" action="gender.php">

    Select Gender:

    <select name="gender">

        <option value="M">Male</option>
        <option value="F">Female</option>

    </select>

    <br><br>

    <input type="submit" value="Display">

</form>

</body>
</html>
```

gender.php

```
<?php

include "db.php";

$gender = $_POST['gender'];

$sql = "SELECT * FROM employee
```

```
        WHERE Gender='$gender'";

$result = mysqli_query($conn, $sql);

echo "<h2>Employee Details</h2>";

echo "<table border='1' cellpadding='10'>";

echo "<tr>";
echo "<th>Emp ID</th>";
echo "<th>Name</th>";
echo "<th>Department</th>";
echo "<th>Joining Date</th>";
echo "<th>Gender</th>";
echo "</tr>";

$count = 0;

while ($row = mysqli_fetch_assoc($result)) {

    echo "<tr>";

    echo "<td>" . $row['Empid'] . "</td>";
    echo "<td>" . $row['Ename'] . "</td>";
    echo "<td>" . $row['Deptcode'] . "</td>";
    echo "<td>" . $row['JoiningDate'] . "</td>";
    echo "<td>" . $row['Gender'] . "</td>";

    echo "</tr>";

    $count++;
}

echo "</table>";

echo "<h3>Total Employees: " . $count . "</h3>";

mysqli_close($conn);

?>
```

9(G) Highlight Newly Inserted Employee

The requirement is to display records **before and after insertion** in separate tables and highlight the inserted employee's row in **pink** in both tables.

insert_highlight.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Employee Insert</title>
</head>
```

```
<body>

<h2>Insert Employee</h2>

<form method="post" action="insert_highlight.php">

    Employee ID:
    <input type="text" name="empid" required>
    <br><br>

    Employee Name:
    <input type="text" name="ename" required>
    <br><br>

    Department:

    <input type="radio" name="deptcode" value="A"> Admin
    <input type="radio" name="deptcode" value="S"> Sales
    <input type="radio" name="deptcode" value="F"> Finance

    <br><br>

    Joining Date:
    <input type="date" name="joiningdate" required>
    <br><br>

    Gender:

    <select name="gender">
        <option value="M">Male</option>
        <option value="F">Female</option>
    </select>

    <br><br>

    <input type="submit" value="Insert">

</form>

</body>
</html>
```

insert_highlight.php

```
<?php

include "db.php";

$empid = $_POST['empid'];
$ename = $_POST['ename'];
$deptcode = $_POST['deptcode'];
$joiningdate = $_POST['joiningdate'];
$gender = $_POST['gender'];

// Get records BEFORE insertion
$before = mysqli_query($conn, "SELECT * FROM employee");
```

```
// Insert new record
$sql = "INSERT INTO employee
        VALUES
        ('$empid','$ename','$deptcode','$joiningdate','$gender')";

mysqli_query($conn, $sql);

// Display BEFORE table
echo "<h2>Records Before Inserting</h2>";

echo "<table border='1' cellpadding='10'>";

echo "<tr>
        <th>Emp ID</th>
        <th>Name</th>
        <th>Department</th>
        <th>Joining Date</th>
        <th>Gender</th>
      </tr>";

while ($row = mysqli_fetch_assoc($before)) {

    echo "<tr>";

    echo "<td>" . $row['Empid'] . "</td>";
    echo "<td>" . $row['Ename'] . "</td>";
    echo "<td>" . $row['Deptcode'] . "</td>";
    echo "<td>" . $row['JoiningDate'] . "</td>";
    echo "<td>" . $row['Gender'] . "</td>";

    echo "</tr>";
}

// Highlight newly inserted employee
echo "<tr style='background-color:pink'>";

echo "<td>" . $empid . "</td>";
echo "<td>" . $ename . "</td>";
echo "<td>" . $deptcode . "</td>";
echo "<td>" . $joiningdate . "</td>";
echo "<td>" . $gender . "</td>";

echo "</tr>";

echo "</table>";


// Display AFTER table

echo "<h2>Records After Inserting</h2>";

$after = mysqli_query($conn, "SELECT * FROM employee");

echo "<table border='1' cellpadding='10'>";

echo "<tr>
```

```
        <th>Emp ID</th>
        <th>Name</th>
        <th>Department</th>
        <th>Joining Date</th>
        <th>Gender</th>
      </tr>";

while ($row = mysqli_fetch_assoc($after)) {

    if ($row['Empid'] == $empid) {

        echo "<tr style='background-color:pink'>";

    } else {

        echo "<tr>";

    }

    echo "<td>" . $row['Empid'] . "</td>";
    echo "<td>" . $row['Ename'] . "</td>";
    echo "<td>" . $row['Deptcode'] . "</td>";
    echo "<td>" . $row['JoiningDate'] . "</td>";
    echo "<td>" . $row['Gender'] . "</td>";

    echo "</tr>";
}

echo "</table>";

mysqli_close($conn);

?>
```

9(H) Insert If Not Existing, Otherwise Display Existing Record

The requirement is to insert a record when it does not exist; otherwise display the existing employee's details.

insert_or_display.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Employee Record</title>
</head>
<body>

<h2>Employee Details</h2>
```

```
<form method="post" action="insert_or_display.php">

    Employee ID:
    <input type="text" name="empid" required>
    <br><br>

    Employee Name:
    <input type="text" name="ename" required>
    <br><br>

    Department:

    <input type="radio" name="deptcode" value="A"> Admin
    <input type="radio" name="deptcode" value="S"> Sales
    <input type="radio" name="deptcode" value="F"> Finance

    <br><br>

    Joining Date:
    <input type="date" name="joiningdate" required>
    <br><br>

    Gender:

    <select name="gender">

        <option value="M">Male</option>
        <option value="F">Female</option>

    </select>

    <br><br>

    <input type="submit" value="Submit">

</form>

</body>
</html>
```

insert_or_display.php

```
<?php

include "db.php";

$empid = $_POST['empid'];
$ename = $_POST['ename'];
$deptcode = $_POST['deptcode'];
$joiningdate = $_POST['joiningdate'];
$gender = $_POST['gender'];

$sql = "SELECT * FROM employee
        WHERE Empid='$empid'";

$result = mysqli_query($conn, $sql);
```

```
if (mysqli_num_rows($result) > 0) {

    // Existing record
    $row = mysqli_fetch_assoc($result);

    echo "<h2>Record Already Exists</h2>";

    echo "<table border='1' cellpadding='10'>";

    echo "<tr>";
    echo "<th>Employee ID</th>";
    echo "<th>Name</th>";
    echo "<th>Department</th>";
    echo "<th>Joining Date</th>";
    echo "<th>Gender</th>";
    echo "</tr>";

    echo "<tr>";

    echo "<td>" . $row['Empid'] . "</td>";
    echo "<td>" . $row['Ename'] . "</td>";
    echo "<td>" . $row['Deptcode'] . "</td>";
    echo "<td>" . $row['JoiningDate'] . "</td>";
    echo "<td>" . $row['Gender'] . "</td>";

    echo "</tr>";

    echo "</table>";

} else {

    // Insert new record
    $sql = "INSERT INTO employee
            VALUES
            ('$empid','$ename','$deptcode',
             '$joiningdate','$gender')";

    if (mysqli_query($conn, $sql)) {

        echo "<h3>New employee record inserted successfully.</h3>";

    } else {

        echo "Error: " . mysqli_error($conn);

    }

}

mysqli_close($conn);

?>
```

9(I) Separate Male and Female Tables

The question requires two separate tables—one for male employees and another for female employees—with summary counts.

gender_tables.php

```
<?php

include "db.php";

// Male employees
$male = mysqli_query(
    $conn,
    "SELECT * FROM employee WHERE Gender='M'"
);

echo "<h2>Male Employees</h2>";

echo "<table border='1' cellpadding='10'>";

echo "<tr>
        <th>Employee ID</th>
        <th>Name</th>
        <th>Department</th>
        <th>Joining Date</th>
        <th>Gender</th>
      </tr>";

$maleCount = 0;

while ($row = mysqli_fetch_assoc($male)) {

    echo "<tr>";

    echo "<td>" . $row['Empid'] . "</td>";
    echo "<td>" . $row['Ename'] . "</td>";
    echo "<td>" . $row['Deptcode'] . "</td>";
    echo "<td>" . $row['JoiningDate'] . "</td>";
    echo "<td>" . $row['Gender'] . "</td>";

    echo "</tr>";

    $maleCount++;
}

echo "</table>";

echo "<h3>Total Male Employees: " .
     $maleCount . "</h3>";


// Female employees
$female = mysqli_query(
```

```
    $conn,
    "SELECT * FROM employee WHERE Gender='F'"
);

echo "<h2>Female Employees</h2>";

echo "<table border='1' cellpadding='10'>";

echo "<tr>
        <th>Employee ID</th>
        <th>Name</th>
        <th>Department</th>
        <th>Joining Date</th>
        <th>Gender</th>
      </tr>";

$femaleCount = 0;

while ($row = mysqli_fetch_assoc($female)) {

    echo "<tr>";

    echo "<td>" . $row['Empid'] . "</td>";
    echo "<td>" . $row['Ename'] . "</td>";
    echo "<td>" . $row['Deptcode'] . "</td>";
    echo "<td>" . $row['JoiningDate'] . "</td>";
    echo "<td>" . $row['Gender'] . "</td>";

    echo "</tr>";

    $femaleCount++;
}

echo "</table>";

echo "<h3>Total Female Employees: " .
     $femaleCount . "</h3>";


// Summary
echo "<h2>Summary</h2>";

echo "Male Employees: " . $maleCount . "<br>";
echo "Female Employees: " . $femaleCount . "<br>";
echo "Total Employees: " .
     ($maleCount + $femaleCount);

mysqli_close($conn);

?>
```

9(J) Fetch Employee into Form and Update

The question requires entering employee ID, fetching the employee record into an HTML form, allowing changes except Employee ID, and updating the database.

edit.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Edit Employee</title>
</head>
<body>

<h2>Search Employee</h2>

<form method="post" action="edit.php">

    Employee ID:
    <input type="text" name="empid" required>

    <br><br>

    <input type="submit" value="Search">

</form>

</body>
</html>
```

edit.php

```
<?php

include "db.php";

$empid = $_POST['empid'];

$sql = "SELECT * FROM employee
        WHERE Empid='$empid'";

$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {

    $row = mysqli_fetch_assoc($result);

?>

<h2>Edit Employee Details</h2>

<form method="post" action="save.php">

    Employee ID:
```

```
<input type="text"
       name="empid"
       value="<?php echo $row['Empid']; ?>"
       readonly>

<br><br>

Employee Name:

<input type="text"
       name="ename"
       value="<?php echo $row['Ename']; ?>">

<br><br>

Department:

<input type="radio"
       name="deptcode"
       value="A"
       <?php if ($row['Deptcode'] == 'A') echo "checked"; ?>>
Admin

<input type="radio"
       name="deptcode"
       value="S"
       <?php if ($row['Deptcode'] == 'S') echo "checked"; ?>>
Sales

<input type="radio"
       name="deptcode"
       value="F"
       <?php if ($row['Deptcode'] == 'F') echo "checked"; ?>>
Finance

<br><br>

Joining Date:

<input type="date"
       name="joiningdate"
       value="<?php echo $row['JoiningDate']; ?>">

<br><br>

Gender:

<select name="gender">

    <option value="M"
    <?php if ($row['Gender'] == 'M') echo "selected"; ?>>
    Male
    </option>

    <option value="F"
    <?php if ($row['Gender'] == 'F') echo "selected"; ?>>
```

```
        Female
        </option>

    </select>

    <br><br>

    <input type="submit" value="Update">

</form>

<?php

} else {

    echo "<h3>Employee record not found.</h3>";

}

mysqli_close($conn);

?>
```

save.php

```
<?php

include "db.php";

$empid = $_POST['empid'];
$ename = $_POST['ename'];
$deptcode = $_POST['deptcode'];
$joiningdate = $_POST['joiningdate'];
$gender = $_POST['gender'];

$sql = "UPDATE employee SET

        Ename='$ename',
        Deptcode='$deptcode',
        JoiningDate='$joiningdate',
        Gender='$gender'

        WHERE Empid='$empid'";

if (mysqli_query($conn, $sql)) {

    echo "<h3>Employee record updated successfully.</h3>";

} else {

    echo "Error: " . mysqli_error($conn);

}

mysqli_close($conn);
```

```
?>
```

9(K) Fetch Employee and Delete

The journal requires entering Employee ID, displaying the employee record, providing a Delete button, and deleting the record when the button is clicked.

delete.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Delete Employee</title>
</head>
<body>

<h2>Delete Employee</h2>

<form method="post" action="delete.php">

    Employee ID:
    <input type="text" name="empid" required>

    <br><br>

    <input type="submit" value="Search">

</form>

</body>
</html>
```

delete.php

```
<?php

include "db.php";

$empid = $_POST['empid'];

$sql = "SELECT * FROM employee
        WHERE Empid='$empid'";

$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {

    $row = mysqli_fetch_assoc($result);

    echo "<h2>Employee Details</h2>";

    echo "<table border='1' cellpadding='10'>";
```

```
    echo "<tr>";
    echo "<th>Employee ID</th>";
    echo "<th>Name</th>";
    echo "<th>Department</th>";
    echo "<th>Joining Date</th>";
    echo "<th>Gender</th>";
    echo "</tr>";

    echo "<tr>";

    echo "<td>" . $row['Empid'] . "</td>";
    echo "<td>" . $row['Ename'] . "</td>";
    echo "<td>" . $row['Deptcode'] . "</td>";
    echo "<td>" . $row['JoiningDate'] . "</td>";
    echo "<td>" . $row['Gender'] . "</td>";

    echo "</tr>";

    echo "</table>";

    echo "<br>";

    echo "<form method='post' action='confirm_delete.php'>";

    echo "<input type='hidden'
                name='empid'
                value='" . $row['Empid'] . "'>";

    echo "<input type='submit'
                value='Delete Employee'>";

    echo "</form>";

} else {

    echo "<h3>Employee record not found.</h3>";

}

mysqli_close($conn);

?>
```

confirm_delete.php

```
<?php

include "db.php";

$empid = $_POST['empid'];

$sql = "DELETE FROM employee
        WHERE Empid='$empid'";

if (mysqli_query($conn, $sql)) {
```

```
    echo "<h2>Employee record deleted successfully.</h2>";
} else {
    echo "Error: " . mysqli_error($conn);
}
mysqli_close($conn);
?>
```

Recommended Folder Structure

For easy practical execution in **XAMPP**, arrange the files like this:

```
C:\xampp\htdocs\
│
├── program1\
│    ├── index.html
│    └── temperature.php
│
├── program2\
│    ├── index.html
│    └── birthdate.php
│
├── program3\
│    ├── index.html
│    └── college.php
│
├── program4\
│    ├── index.html
│    ├── first.php
│    └── second.php
│
├── program5\
│    ├── index.html
│    ├── first.php
│    └── second.php
│
├── program6\
│    ├── index.html
│    ├── first.php
│    └── log.txt
│
├── program7\
│    ├── index.html
│    └── upload.php
│
├── program8\
│    └── student.php
│
```

```
├── program9\
│   ├── db.php
│   ├── insert.html
│   ├── insert.php
│   ├── check.html
│   ├── check.php
│   ├── update.html
│   ├── update.php
│   ├── firstrecord.php
│   ├── firsttable.php
│   ├── gender.html
│   ├── gender.php
│   ├── insert_highlight.html
│   ├── insert_highlight.php
│   ├── insert_or_display.html
│   ├── insert_or_display.php
│   ├── gender_tables.php
│   ├── edit.html
│   ├── edit.php
│   ├── save.php
│   ├── delete.html
│   ├── delete.php
│   └── confirm_delete.php
│
└── images\
```

How to Run

Start **Apache** and **MySQL** in XAMPP.

Then open:

```
http://localhost/program1/index.html
```

For example, Program 9:

```
http://localhost/program9/insert.html
```

For database programs, first create the **LYBCA** database and **employee** table given above.